



ALMA MATER STUDIORUM  
UNIVERSITÀ DI BOLOGNA

# Statistical inference of transcriptomics data

## Part V: proteomics

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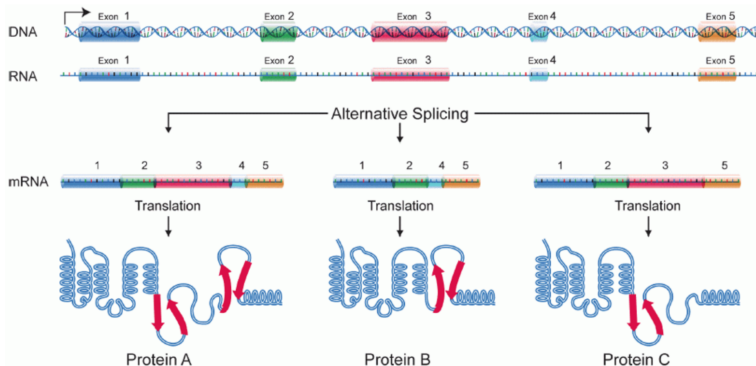
Proteomics (LC-MS data)

Differential testing (LC-MS data) - Ongoing work

Protein interactions - Ongoing work

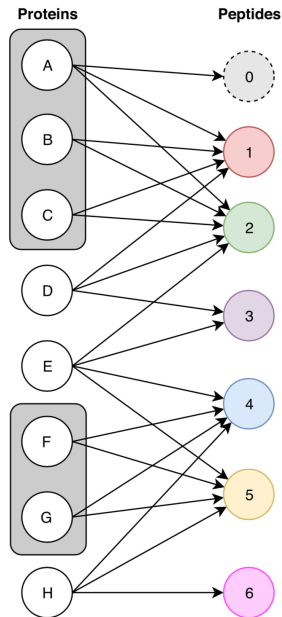
Summary

# Proteomics



- Proteomics focuses on protein abundance.
- Analyses are similar to those seen for transcriptomics.
- Proteins are biologically more interesting than mRNA.
- However, proteomics data is a lot more noisy, which is why most analyses are done on mRNA.

- In “bottom-up” proteomics, peptides are used as surrogate markers for proteins.
- Major technical issues:
  - ▶ most peptides can originate from multiple proteins;
  - ▶ peptides may be erroneously detected, even when absent.
- In our data: 71% of abundance is shared across isoforms, and 31% is shared across genes.
- Our aim: go from peptide- to protein-level data.
- If available, we enhance data by using mRNA data (correlated and more accurate).



- Observed data:
  - ▶  $Y = (Y_1, \dots, Y_N)$ , peptide **abundance** (PSM counts or intensity);
  - ▶  $\psi = (\psi_1, \dots, \psi_N)$ , protein-peptide mapping (**mapping**);
  - ▶  $PEP = (PEP_1, \dots, PEP_N)$ , **probability** of peptide erroneous detection;
  - ▶  $\pi^{(T)} = (\pi_1^{(T)}, \dots, \pi_P^{(T)})$ , RNA-seq relative abundance (**informative prior**).
- Unknown parameters:
  - ▶  $X = (X_1, \dots, X_P)$ , protein-isoform **abundance**;
  - ▶  $\pi = (\pi_1, \dots, \pi_P)$ , protein-isoform **relative abundance**.
- Aim: recover  $(X, \pi)$  starting from  $(Y, \psi, PEP, \pi^{(T)})$ .

$$X|\pi \sim \text{Multinom}\left(\pi, n = \sum_{p=1}^P X_p\right) \quad (1)$$

$$\pi \sim \text{Dirichlet}(\delta), \quad (2)$$

where  $\delta \propto \pi^{(T)}$ , our **informative prior** from mRNA data (if available).

- Posterior (due to conjugate prior):

$$\pi|X = x, \delta \sim \text{Dirichlet}(x_1 + \delta_1, \dots, x_P + \delta_P). \quad (3)$$

- $X$  is unknown: we recover it via a **2-layer latent variable model**.

- ▶ 1<sup>st</sup> layer - peptide misidentification ( $\epsilon_i = 1$  or 0):

$$\epsilon_i|PEP_i \sim \text{Bern}(PEP_i), \text{ for } i = 1, \dots, N. \quad (4)$$

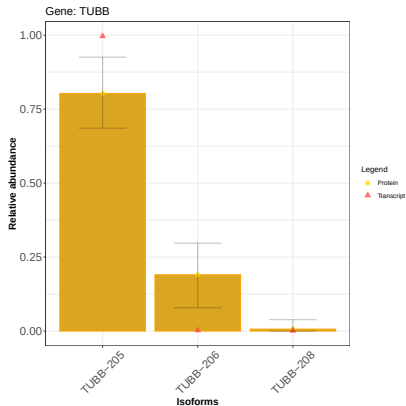
- ▶ 2<sup>nd</sup> layer - allocate peptide abundance (if  $\epsilon_i = 0$ ):

$$(X_{1i}, \dots, X_{Ni})|\pi, \psi_i, Y_i, \epsilon_i \sim \text{Multinom}\left(\tilde{\pi}^{(i)}, Y_i(1 - \epsilon_i)\right), \quad (5)$$

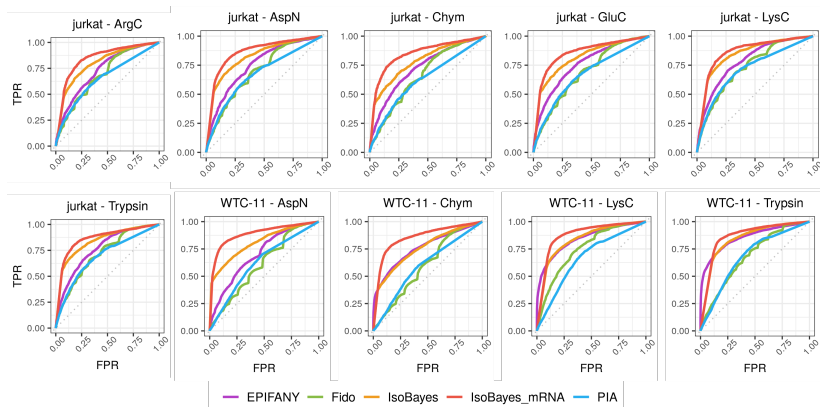
for  $i = 1, \dots, N$ , where  $\tilde{\pi}_p^{(i)} \propto \pi_p \mathbb{1}(p \in \psi_i)$ .

- Inference via three Gibbs samplers:
  - ▶  $\pi|X, \delta \sim \text{Dirichlet}(X + \delta)$  (Gibbs);
  - ▶  $\epsilon|PEP \sim \text{Bern}(PEP)$  (Gibbs);
  - ▶  $X|\pi, \psi, Y, \epsilon$  via several Multinomial distributions (1 per shared peptide) (Gibbs).

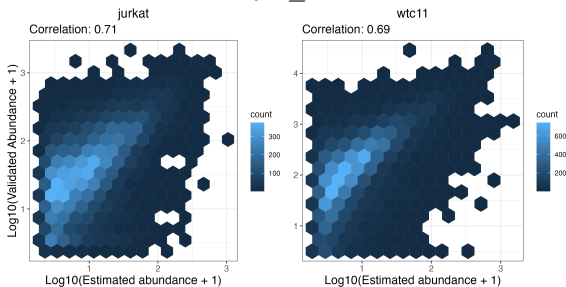
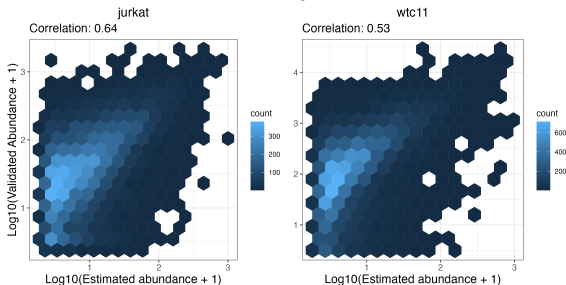
| Gene  | Isoform   | Prob_present | Abundance | CI_LB | CI_UB |
|-------|-----------|--------------|-----------|-------|-------|
| AARS1 | AARS1-201 | 0.997        | 31.980    | 18    | 34    |
| AARS1 | AARS1-203 | 0.116        | 1.518     | 0     | 12    |
| AARS1 | AARS1-217 | 0.039        | 0.317     | 0     | 0     |
| AARS1 | AARS1-231 | 0.042        | 0.181     | 0     | 0     |
| TUBB  | TUBB-205  | 1.000        | 69.471    | 61    | 77    |
| TUBB  | TUBB-206  | 1.000        | 13.956    | 9     | 20    |
| TUBB  | TUBB-208  | 0.052        | 0.169     | 0     | 1     |

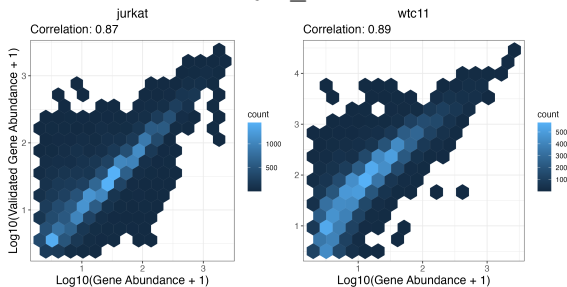
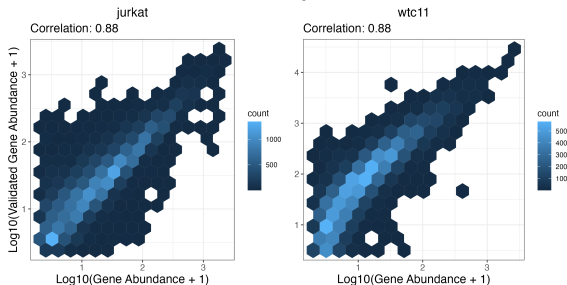


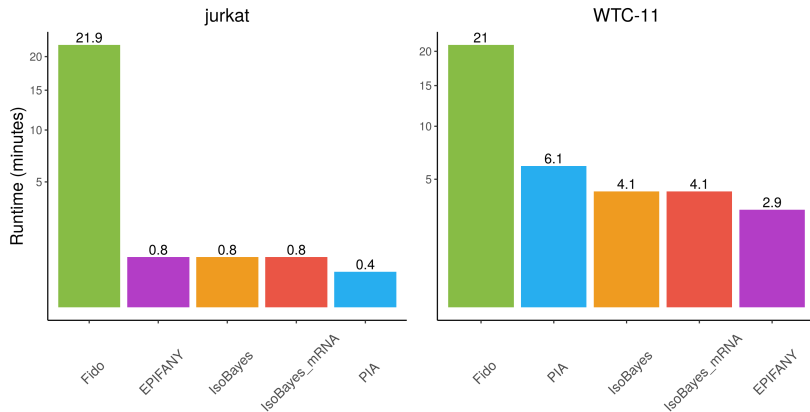
- Initial validation in simulations.
- Hard to reproduce the complexity of the measurement process in simulations.
- Validation on real data (Sheynkman lab), 2 multi-protease datasets.
- Cross-validation approach: analyze 1 protease; use the remaining ones to build a (noisy) ground truth (from the uniquely mapping peptides).
- Methods benchmarked:
  - ▶ *IsoBayes* with mRNA abundances (*IsoBayes\_mRNA*);
  - ▶ *IsoBayes* without mRNA abundances;
  - ▶ EPIFANY;
  - ▶ Fido;
  - ▶ PIA.
- Validate:
  - ▶ presence/absence of protein isoforms (ROC curve);
  - ▶ abundance of protein isoforms (log10 correlation).
- N.B. Competitors cannot incorporate mRNA data, and do not provide abundance information.



ROC curve for the detection of protein isoforms ( $Pr(X_p > 0)$ ).

*IsoBayes\_mRNA**IsoBayes*

*IsoBayes\_mRNA**IsoBayes*



Average runtime (in minutes) per dataset (8 cores available).  
Coding techniques: C++ and parallel coding.

Proteomics (LC-MS data)

Differential testing (LC-MS data) - Ongoing work

Protein interactions - Ongoing work

Summary

- Hierarchical extension (sharing of information across samples) and differential testing between conditions:
  - ▶ gene-level differential abundance (DA);
  - ▶ differential alternative splicing (DAS) or differential isoform usage (DIU) within genes (gene- and isoform-level): first tool for protein-level DAS.

- Two hierarchical models across  $N$  samples.
- Gene-level counts ( $G$  genes) for sample  $i$ ,  $X^{(i)}$ :

$$X^{(i)} | \pi^{(i)}, \sim \text{Multinom} \left( \pi^{(i)}, n = \sum_{g=1}^G X_g^{(i)} \right) \quad (6)$$

$$\pi^{(i)} = \left( \pi_1^{(i)}, \dots, \pi_G^{(i)} \right) \sim \text{Dirichlet}(\delta), \quad (7)$$

- Isoform-level counts ( $K_g$  isoforms) for sample  $i$  within gene  $g$ ,  $Y_g^{(i)}$ :

$$Y_g^{(i)} | \pi^{(i)}, X^{(i)} \sim \text{Multinom} \left( \omega_g^{(i)}, n = X_g^{(i)} \right) \quad (8)$$

$$\omega_g^{(i)} = \left( \omega_{g1}^{(i)}, \dots, \omega_{gK_g}^{(i)} \right) \sim \text{Dirichlet}(\alpha_g), \quad (9)$$

- Sampling scheme in 5 steps:
- Gene-level parameters:
  - ▶  $\delta | \pi^{(1)}, \dots, \pi^{(N)}$  (Metropolis-Hasting);
  - ▶ for  $i = 1, \dots, N$ :  
 $\pi^{(i)} | \delta, X^{(i)} \sim \text{Dirichlet}(\delta + X^{(i)})$  (Gibbs);
- Isoform-level parameters, for  $g = 1, \dots, G$ :
  - ▶  $\alpha_g | \omega_g^{(1)}, \dots, \omega_g^{(N)}$  (Metropolis-Hasting);
  - ▶ for  $i = 1, \dots, N$ :  
 $\omega_g^{(i)} | \alpha_g, Y_g^{(i)}, X^{(i)} \sim \text{Dirichlet}(\delta + X^{(i)})$  (Gibbs);
- Latent variables, for  $i = 1, \dots, N$ :
  - ▶  $X^{(i)}, Y^{(i)} | \pi^{(i)}, \omega_1^{(i)}, \dots, \omega_G^{(i)}$  via several Multinomial distributions (1 per shared peptide) (Gibbs).

- Consider two conditions,  $A$  and  $B$ .
- DGE, compare  $\bar{\delta}_g = \frac{\delta_g}{\sum_{g'=1}^G \delta_{g'}}$ , for gene  $g$ :

$$H_0 : \bar{\delta}_g^A = \bar{\delta}_g^B \quad (10)$$

$$H_1 : \text{otherwise.} \quad (11)$$

- Gene-level DAS, compare all  $\bar{\alpha}_{gk} = \frac{\alpha_{gk}}{\sum_{k'=1}^{K_g} \alpha_{gk'}}$ , for gene  $g$ :

$$H_0 : \bar{\alpha}_{gk}^A = \bar{\alpha}_{gk}^B, \text{ for } k = 1, \dots, N_k \quad (12)$$

$$H_1 : \text{otherwise.} \quad (13)$$

- Isoform-level DAS, compare individual  $\bar{\alpha}_{gk}$ , for isoform  $k$  in gene  $g$ :

$$H_0 : \bar{\alpha}_{gk}^A = \bar{\alpha}_{gk}^B \quad (14)$$

$$H_1 : \text{otherwise.} \quad (15)$$

**DA:**

| Gene    | Pi_A  | Pi_B  | Pr_greater | log2FC  |
|---------|-------|-------|------------|---------|
| CDC26   | 3e-04 | 0e+00 | 1          | 4.0385  |
| SCAF1   | 4e-04 | 0e+00 | 1          | 3.8839  |
| C1GALT1 | 0e+00 | 1e-04 | 0          | -3.7153 |
| CD247   | 6e-04 | 0e+00 | 1          | 3.6494  |
| SEC23IP | 1e-03 | 1e-04 | 1          | 3.6161  |
| MT-ATP8 | 1e-04 | 0e+00 | 1          | 3.5796  |

**DAS (or DIU):**

| Isoform   | Gene  | gene_FDR | Pi_A   | Pi_B   | Pr_greater | log2FC  |
|-----------|-------|----------|--------|--------|------------|---------|
| ACIN1-223 | ACIN1 | 0        | 0.0613 | 0.7136 | 0.0000     | -3.6900 |
| ACIN1-217 | ACIN1 | 0        | 0.0869 | 0.0283 | 0.9652     | 1.7723  |
| ACIN1-201 | ACIN1 | 0        | 0.4398 | 0.0463 | 0.9436     | 3.0720  |
| ACIN1-203 | ACIN1 | 0        | 0.0564 | 0.1184 | 0.1062     | -1.1444 |
| ACIN1-205 | ACIN1 | 0        | 0.3026 | 0.0468 | 0.8656     | 2.2514  |
| ACIN1-222 | ACIN1 | 0        | 0.0531 | 0.0466 | 0.5886     | 0.3232  |
| ACLY-201  | ACLY  | 0        | 0.0676 | 0.8361 | 0.0000     | -3.8386 |
| ACLY-204  | ACLY  | 0        | 0.7723 | 0.0920 | 1.0000     | 3.3203  |
| ACLY-202  | ACLY  | 0        | 0.1601 | 0.0720 | 0.8816     | 1.4093  |

Proteomics (LC-MS data)

Differential testing (LC-MS data) - Ongoing work

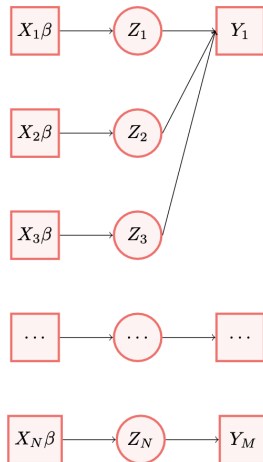
Protein interactions - Ongoing work

Summary

- Gene-level protein interactions available (estimated).
- STRING database: a score  $[0, 1,000]$  for each pair of genes (we threshold at 700).
- Isoform-level interactions currently unknown!
- Assumption: if two genes interact, at least 1 pair of the respective isoforms must interact.
- For each gene pair: about 35 isoform pairs.
- Aim: for each interacting gene pair, estimate the probability that each isoform pair interacts (based on STRING data).

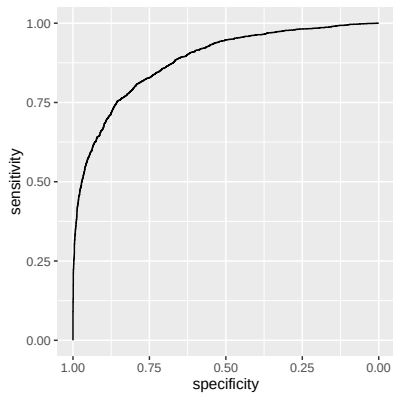
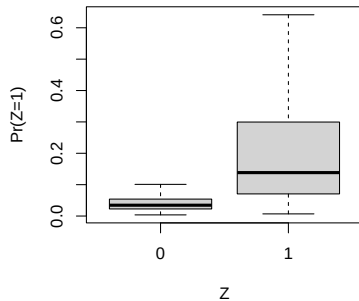
- For each isoform, 2 pieces of information:
  - ▶ RNA-seq TPMs from 54,383 samples;
  - ▶ 1,152 embeddings calculated from the protein sequence (via ESM Cambrian tool).
- For each isoform pair (82 covariates):
  - ▶ absolute (Pearson) correlation of TPMs (co-expression of transcripts);
  - ▶ (Pearson) correlation of embeddings (similarity of embeddings);
  - ▶ PCA of embeddings (top 80 components, 80% of variance explained): we add the components of the two isoforms.

- Challenge: inputs at isoform-level, while outputs at the gene-level!
- 3 layers:
  - ▶ iso-pair level inputs  $X$ ;
  - ▶ iso-pair level interactions  $Z$ ;
  - ▶ gene-pair level interactions  $Y$ .
- 2 connections:
  - ▶  $Pr(Z_i = 1) = \text{invprobit}(X_i\beta)$  (probit model);
  - ▶  $Y_g = \max_{i \in \psi_g}(Z_i)$   
( $\psi_g$  = list of isoform pairs for gene pair  $g$ ).
- Bayesian approach, MCMC sampling:
  - ▶  $\beta|Z$  (Gibbs; Holmes and Held 2006);
  - ▶  $Z|\beta, Y$  (Metropolis Hasting).



- Computational issue: 20,353 genes = 207 million pairs of genes; 93,521 isoforms = 4.4 billion isoform pairs.
- We take all  $\sim 230$  k genes with  $\text{STRING} \geq 700$ , and an equal number of controls; which leads to  $\sim 461$  k gene pairs, and  $\sim 13$  m iso pairs.
- Full algorithm runs in  $\sim 2$  h on 1 core.
- Ongoing validation of results.

- Initial validation in a simulation study:  $AUC > 0.99$ .
- Real data validation via a test-set approach: 50% genes in train and 50% in test.
- Train set: pairs of genes in train set (25% of data); test set: pairs of genes in test set (25% of data).
- Infer  $\hat{\beta}$  from train set, and estimate  $\hat{\pi}_i = \hat{Pr}(Z_i = 1) = \text{invprobit}(X_i \hat{\beta})$  on test set.
- Compare  $\hat{\pi}_i$  with subset of known  $Z_i$ : genes pairs with 1 isoform only.
- Most recent result:  $AUC = 0.95$ .

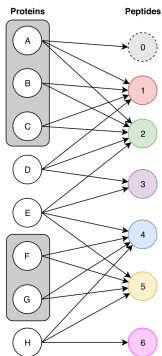


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